

CSE 232: Introduction to Programming II

Summer 2022 (US22)

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College of Engineering
Department of Computer Science & Engineering

Introduction

■ *Course Description*

- Continuation of *object-oriented design* and *implementation in C++*
- Building *programs from modules*
- *Data abstraction* and classes to implement *abstract data types*
- Static and dynamic *memory allocation*
- *Data structure* implementation and *algorithm efficiency*
- *Lists, tables, stacks, and queues*
- *Templates* and *generic programming*

Introduction

■ *Course Expectations*

- Prior programming experience
- We will be using the latest C++ standard (**C++20**)
- We will focus on the C++ *Standard Template Library (STL)*

This is where you learn C++.



The World Map of C++ STL Algorithms

[\(https://www.fluentcpp.com/2018/07/06/world-map-cpp-stl-algorithms/\)](https://www.fluentcpp.com/2018/07/06/world-map-cpp-stl-algorithms/)

Introduction

■ *Course Instructor*

- *Dr. Michael Langford*
- *Ph.D. Computer Science*
 - Systems for Trusted AI / Robustness & Resiliency
- *Prior experience*
 - Software Industry, Military (Navy)
- *Areas of interest*
 - Software Engineering for AI, Autonomous Systems
 - Deep Learning, Evolutionary Computation



Dr. Michael Langford
(Course Instructor)

Hello, I am your instructor.



Asynchronous Learning

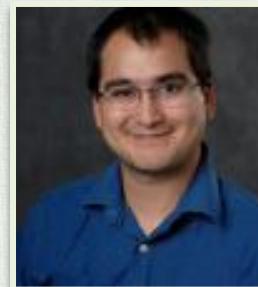
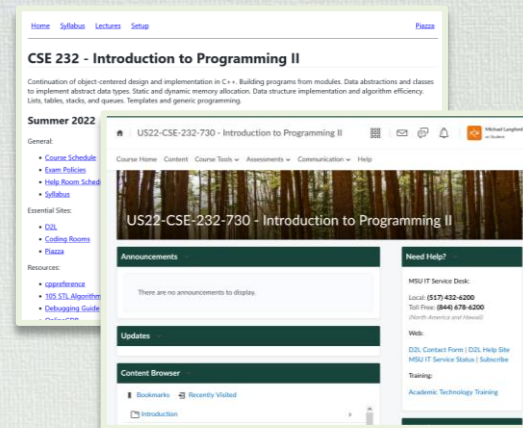
■ Lectures

- Lectures available as *pre-recorded videos* (with slides)
- Website: <https://cse.msu.edu/~cse232/>
- D2L: <https://d2l.msu.edu> (US22-CSE-232-730)
- Lectures are *also available on D2L* under the “Content” header

■ Previously-Recorded

- All lectures have been previously recorded by *Dr. Joshua Nahum*
- *Dr. Nahum is not the instructor of the course this semester*
- *All questions/concerns* about the course should be *directed to Dr. Langford*

We are *leveraging Dr. Nahum's lectures*, but he is not the instructor for this session.



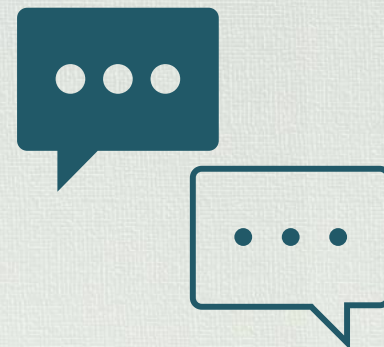
Dr. Joshua Nahum

Communications



■ *Piazza*

- All communication will be handled through *Piazza*
- Link: <https://piazza.com/msu/summer2022/cse232>
- *Do not email instructor or TAs directly*
- Instead, message “Instructors” through Piazza
- Any *questions containing code/answers* should be *posted privately*

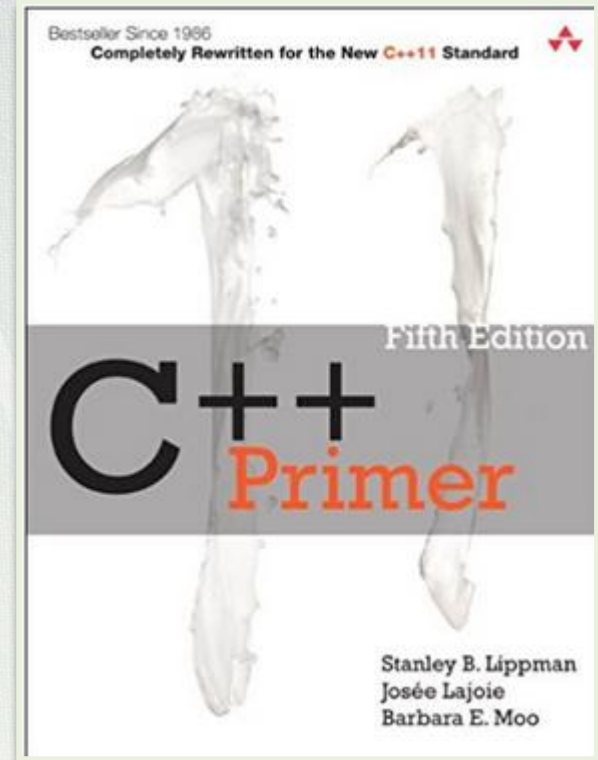


Direct all your questions/concerns through Piazza.

Recommended Textbook

- ***C++ Primer, 5th ed.***
 - by Lippman, Lajoie, and Moo
 - *This is a very useful reference*
 - **Note:** textbook is optional (not mandatory)
 - Highly recommended if you prefer learning by reading

This textbook is optional but highly recommended.



Asking for Help

■ *Help Rooms*

- Private *one-on-one assistance* will be provided through *help rooms over Zoom*
- Attendees will be placed in individual breakout rooms and *assisted on a first-come basis*
- Course staff will join your breakout room *when it is your turn*
- *Zoom session details are provided on Piazza*

One-on-one assistance with the instructor and TAs will be *managed through Zoom.*

Assignments



Coding Rooms

■ **Coding Rooms**

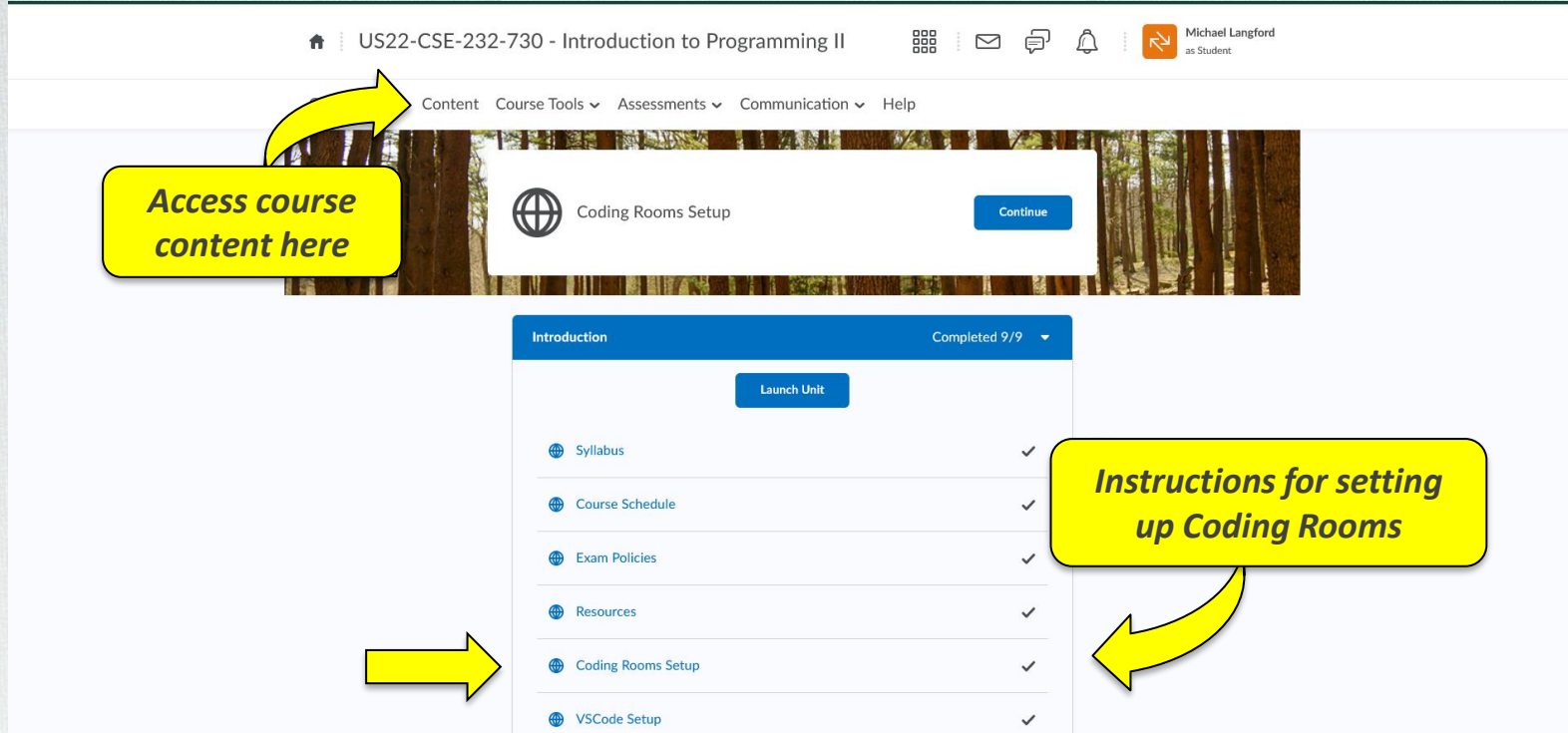
- **All assignments** will be submitted and graded using the **Coding Rooms** website
- A major benefit: enables you to **check your solution before you submit** your work
- Unfortunately, it **requires a payment of \$15**
- Your payment will be refunded if you drop the class prior to the Add/Drop date

■ **Signing up**

- You can **sign up for coding rooms through D2L**
- Go to **Content > Introduction > Coding Rooms Setup** for more information
- Closely follow the instructions to ensure you enroll properly

Coding Rooms is mandatory for this course. It is highly recommended that you sign up through D2L (**see the following slides**)

Coding Rooms



US22-CSE-232-730 - Introduction to Programming II

Content Course Tools ▾ Assessments ▾ Communication ▾ Help

Access course content here

 Coding Rooms Setup [Continue](#)

Introduction Completed 9/9 ▾

[Launch Unit](#)

-  Syllabus ✓
-  Course Schedule ✓
-  Exam Policies ✓
-  Resources ✓
-  Coding Rooms Setup ✓
-  VSCode Setup ✓

Instructions for setting up Coding Rooms

Coding Rooms

Back to Content

Coding Rooms Setup

Print
Download

Coding Rooms Setup

Coding Rooms is the grading platform that we use in CSE 232 to submit work, grade your assignments and show examples. It is a required part of the class. Although we wish it was free, it costs \$15. Please note that if your payment isn't processed (which can take up to 5 days) by the Add/Drop deadline, you will not be able to view or submit assignments. This will likely result in you failing those assignments, so please don't delay.

Coding Rooms Course Payment Instructions

IMPORTANT: To ensure an uninterrupted access to our course material on Coding Rooms, you must make a \$15 payment prior to June 1st. Coding Rooms reserves the right to restrict any unpaid access until full payment is made. The unblocking process may take up to 5 business days, and it is highly recommended for you to process payments before the deadline to avoid any interruption.

- Use the following link or QR code to make your payment via Stripe, **but make sure to pay with your school email address** (@msu.edu).

Note: Failure to pay with your @msu.edu account may restrict your ability to access course assignments on the Coding Rooms website.

[Stripe Payment Link]: <https://buy.stripe.com/biY7vt8Ck4mpbw49AP>



Scan to pay
- You should now be able to access course assignments through the "Coding Rooms" unit found on the D2L

Follow these directions to make your payment

Make sure you use your MSU email address!

Coding Rooms

 US22-CSE-232-730 - Introduction to Programming II
 



 Michael Langford
as Student

[Course Home](#)
[Content](#)
[Course Tools](#)
[Assessments](#)
[Communication](#)
[Help](#)

 Coding Rooms Setup
 [Continue](#)

Introduction Completed 9/9 ▶

Coding Rooms Completed 1/1 ▼

[Launch Unit](#)

 Coding Rooms Login ✓

Week 1 Completed 10/10 ▶

Week 2 Completed 12/12 ▶

Week 3 Completed 11/11 ▶

After you make your payment, use this link to access the course material in Coding Rooms

Grading

■ **Grade Composition**

- **2 Exams** (25% final grade)
- **11 Homeworks** (75% final grade)

■ **Exams**

- Exams will be held **synchronously at assigned times**
- See: https://cse.msu.edu/~cse232/us22/exam_policies.html

■ **Homework**

- Due each **Monday** and **Thursday** at **10:00 PM Eastern Time**
- Homework 00 (due the first Thursday) is not worth any points
- Your **lowest scoring assignment** will be **dropped**

Percent	Points	GPA
90-100%	864-960	4.0
85-89%	816-863	3.5
80-84%	768-815	3.0
75-79%	720-767	2.5
70-74%	672-719	2.0
65-69%	624-671	1.5
60-64%	576-623	1.0
0-59%	0-575	0.0

Grading

■ **Auto-grading**

- **Coding Rooms** will *automatically grade* all assignments
- Coding exercises have *test cases to evaluate your code*
- Many test cases will be visible to you, but *some will be hidden*
- *Partial credit will be awarded*, dependent upon which test cases pass
- A few assignments do have partial *manual-grading components*

■ **No Extra Credit**

- There will be *no extra credit points in this course*
- No bonus points, no extra assignments, no credit for attendance
- All points that you can earn are through exams and homework

Course Schedule

■ *Accelerated Course*

- Homework will be due *every Monday and Thursday (at 10PM!)*
- *Do not wait* until last minute
- You will be able to *work ahead*
- *Exams will be fixed* on the assigned dates

No late assignments will be accepted.

Week	Due Monday	Due Thursday
2022-05-16		HW 00
2022-05-23	HW 01	HW 02
2022-05-30	HW 03	HW 04
2022-06-06	HW 05	Mid-Term
2022-06-13	HW 06	HW 07
2022-06-20	HW 08	HW 09
2022-06-27	HW 10	Final

Course Policies

■ **Plagiarism**

- **Do your own work** on all assignments
- **Do not copy** code from anywhere without attribution
- **Do not share** your solutions with others (*you will also be penalized*)
- **Penalties:** zeros on assignments, final grade reduction, course failure
- All instances will be **reported to the university** (*no exceptions*)
- Consult the syllabus concerning **academic dishonesty**

■ **Chegg and Similar Sites Policy**

- Do not base your work solutions found through external sites such as **Chegg, Brainly, Quizlet, etc.**

Chegg®

This is a serious issue.

It is not unusual for students to fail this course when cases of academic dishonesty are discovered.

Essential Sites

■ *Course Resources*

- *Website:* <http://cse.msu.edu/~cse232/>
- *D2L:* <https://d2l.msu.edu> (US22-CSE-232-730)
- *Piazza:* <https://piazza.com/msu/summer2022/cse232>